1. Introduction

House price prediction is an important task in the real estate industry. It helps buyers, sellers, and investors make better decisions. Accurately estimating house prices is great for investment opportunities for both individuals and businesses. This project focuses on building a machine learning model to predict house prices by analyzing various factors, such as location, size, number of rooms, and additional features. By utilizing historical and current data, our goal is to develop a system that can provide accurate predictions, making the process of buying and selling homes more efficient and transparent.



Figure 1: House price prediction image

In this project, two machine learning methods will be used i.e. Linear Regression and Artificial Neural Networks (ANN). Linear Regression is one of the simplest but powerful techniques. It is well suited for understanding linear relationships between variables. It serves as a baseline approach. It is particularly effective for house price prediction due to its simplicity, interpretability, and wide usage in similar problems. In addition, we will also be using ANN as second algorithm to capture more complex patterns in the data which enables the model to handle non-linear relationships and improve accuracy. By using these two approaches, our project aims to compare the models performances, understand their strengths and weaknesses, and determine the most effective method for predicting house prices.

2. Problem Domain

2.1. Problem

Predicting house prices is one of the most important problems in the real estate market. It helps buyers, sellers, and investors make better decisions. In Nepal, the housing market is growing, and factors like location, size, and features of a house greatly affect its price. But there are no easy and reliable tools to estimate house prices, which often leads to confusion and biased pricing. This project aims to solve this problem by creating a machine learning model that can accurately predict house prices based on key features.



Figure 2: House price prediction image 2

2.2. Dataset

The dataset used for this project is the Nepal house dataset from Kaggle. It includes detailed information about houses in Nepal. The features of the dataset are:

S.N.	Column Name	Description
1	Title	Description or title of the property listing.
2	Location	Address of the property.
3	Price	Price of the house in Nepalese Rupees (NRs).
4	Land Area	Total land area, measured (Anna, square feet, square meter etc.)
5	Build Area	Area occupied by the house, measured in square feet.
6	Road Access	Width of the road leading to the property.
7	Facing	The direction the property faces (e.g., East, South).
8	Floor	Number of floors in the building.
9	Bedroom	Number of bedrooms.
10	Bathroom	Number of bathrooms.
11	Build Year	Year of construction (B.S.).
12	Parking	Number of parking spaces available.
13	Amenities	Additional facilities provided with the house.

Table 1: Columns description table

This dataset is provided by Nepal Homes. It gives valuable insights into the factors that affect house prices in Nepal. By analyzing these features, this project aims to create a machine learning model that can accurately predict house prices and help streamline the buying and selling process in the Nepali real estate market.

Link to the Dataset: https://www.kaggle.com/datasets/nishanpokh/house-price-dataset-nepal/data

3. Solution

Our solution to predicting house prices is by building machine learning models using two algorithms: Linear Regression and Artificial Neural Networks (ANN). Linear Regression is chosen for its simplicity and effectiveness in modeling relationships between variables, serving as a baseline for the predictions. ANN is included to capture complex, non-linear patterns in the dataset, enhancing the accuracy of predictions. Finally, most accurate (better performing) algorithm is selected between these and is used for inference.

3.1. Algorithms Used

The two machine learning algorithms used in the project are described below:

3.1.1. Linear Regression

Linear regression is a machine learning technique that tries to find a linear connection between a target value/variable and one or more input values/variables. It does this by drawing a best fitting line that most closely matches the data (GeeksForGeeks, 2024).

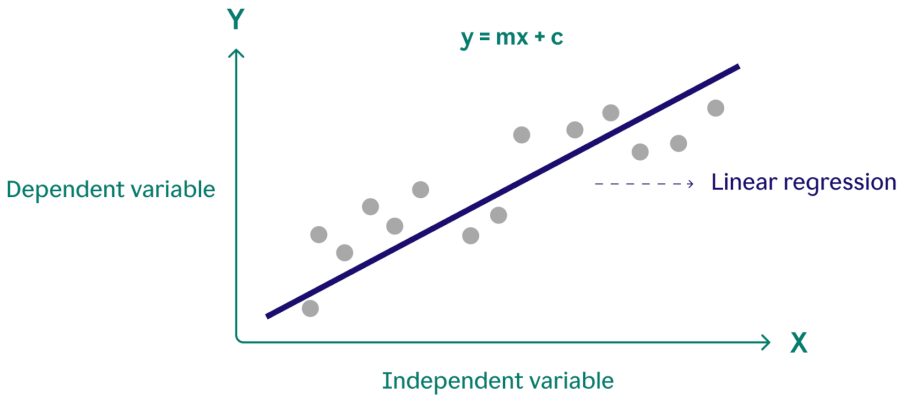


Figure 3: Linear Regression

Reasons for selection:

- Linear relationship between data.
- High correlated features.
- Baseline model for comparison.

3.1.2. Artificial Neural Network (ANN)

A neural network is a machine learning model that makes decisions in a way like the human brain. It works by mimicking how biological neurons interact to recognize patterns, consider different possibilities, and make decisions (IBM, 2024).

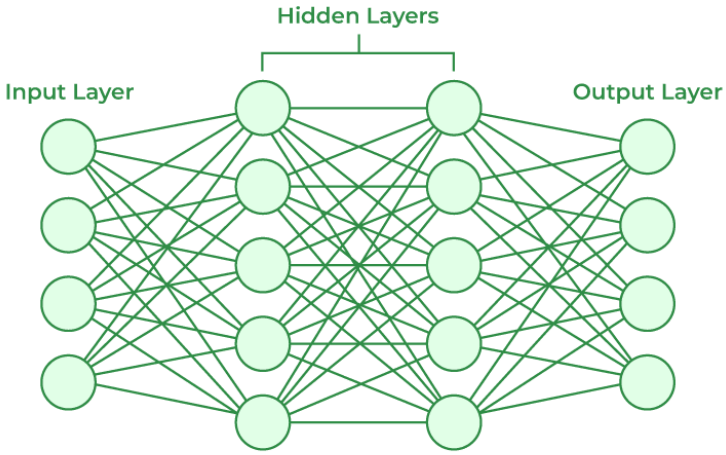


Figure 4 :Artificial Neural Network

Reasons for selection:

- Recognize complex/non-linear patterns.
- Feature Interactions.
- Robustness in noise.

3.2. Flowchart

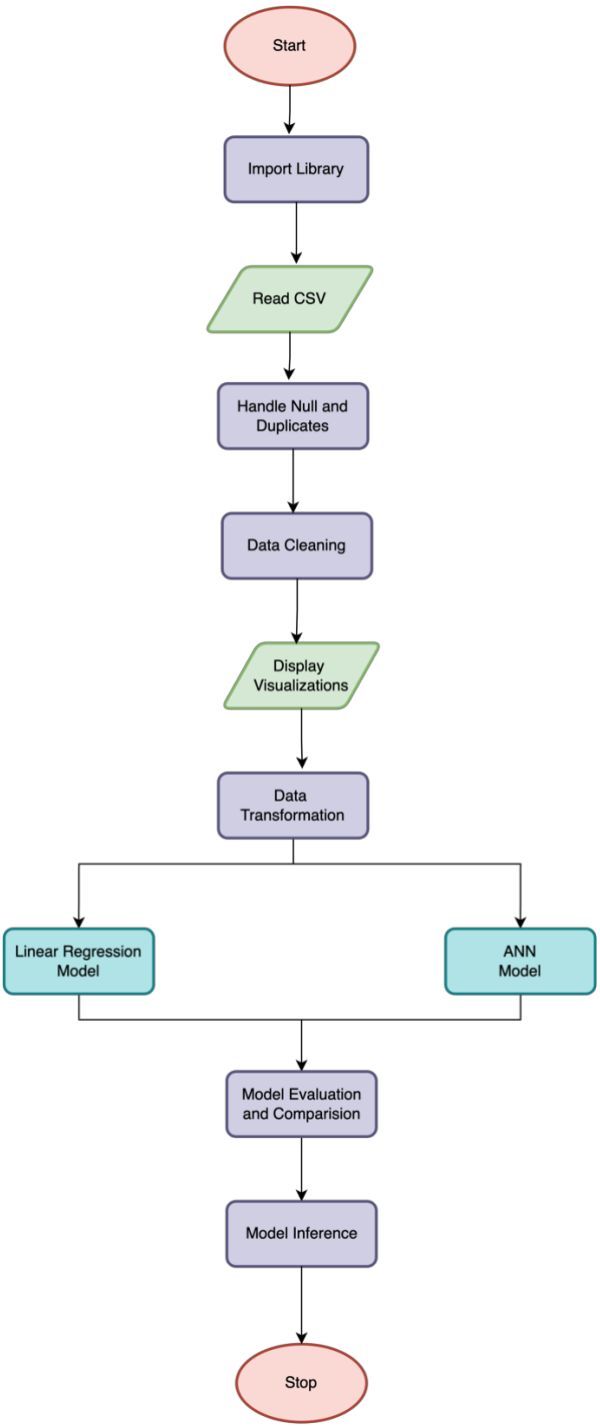


Figure 5: Flowchart of the Project

3.3. Development Process

The following processes are followed step-by-step for the development of this project.

3.3.1. Data Collection

This is the first step for any AI/Machine Learning tasks. I explored various websites, GitHub repositories and Kaggle to find the most suitable dataset that matches the context of Nepal. After deep research, I selected the Nepal House Dataset from Kaggle as the data for this project which was also present in a GitHub repository. This dataset contains columns such as property location, price, land area, build area, road access, number of floors, and amenities. The dataset was provided by Nepal Homes and serves as the foundation for building and evaluating the models.

3.3.2. Data Understanding

Data understanding is the second step of this project. It involves analyzing the dataset to gain a clear understanding of its structure and content. At this step, I read the data, its columns and values, data types, units, statistics, null values, duplicates and so on. This will make us easier for the further processing of the data.

3.3.3. Data Preprocessing

This step involves handling the missing values and duplicates, encoding the categorical columns, converting into same scale and unit. This is a crucial step as it directly determines a model's performance. After this step, the data should be clean, ready and standardized for building models.

3.3.4. Data Visualization

In the fourth step, we visualize the data. Data trends, patterns and relationships are discovered and visualized. Outliers and skewness is also observed in this process. The visualizations include piecharts, bargraphs, histograms, boxplots, scatter plots, correlation heatmap and so on. It will also be a great help for model selection.

3.3.5. Feature Selection

Feature selection is the next step. In this step, the most important features for training the model are identified. This involves selecting correlated and relevant features that have a significant impact on house prices. The selected dataset is then split into training and testing sets, with 80% of the data used for training the model and the remaining 20% for testing its performance. This ensures the model is trained on meaningful data and evaluated effectively. Optionally, the data can also be split for validation during training.

3.3.6. Model Development

Two models were developed i.e. Linear regression model and ANN model. Both models will be developed using the PyTorch library. For both models, Mean Squared Error (MSE) will be used as the loss function to minimize the difference between predicted and actual house prices. The Linear Regression model will provide a simple baseline, while the ANN model will capture more complex, non-linear relationships in the data.

3.3.7. Model Evaluation

Both models are evaluated on the test dataset using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), R-squared (R^2) and Root Mean Squared Error (RMSE). This process helps to evaluate the model's performance and select the best model. Scikit learn library is for the evaluation of models.

3.4. Tools, Technologies, and Libraries Employed

Tools and libraries used throughout the project are explained below:

3.4.1. Data Handling

1. Pandas

Pandas is a python library which mainly is used for managing datasets. It offers various functions for data analysis, cleaning, exploration, and manipulation. This library helps in the analysis of large dataset and get results through statistical methodologies. With Pandas, messy datasets can be cleaned, making them meaningful and relevant (Refsnes Data, 1999-2024).



Figure 6: Pandas logo

3.4.2. Modeling

1. Pytorch

Pytorch is a free and open-source machine learning framework built on the Python programming language and the Torch library. It is an open source library designed for developing deep neural networks. Pytorch has become a popular choice for deep learning research due to its flexibility and efficiency. (Informa TechTarget, 2022).



Figure 7: Pytorch logo

3.4.3. Visualization

1. Matplotlib

Matplotlib is a powerful library in python for making all sorts of graphs and charts. We can create high-quality plots for reports or presentations. It is easy to customize how our graphs looks. It also works well with other tools like JupyterLab, Jupyter Notebook and different graphical interfaces (Matplotlib, 2012-2023).



Figure 8: Matplotlib Logo

2. Seaborn

Seaborn is a Python library for data visualization that is built on top of Matplotlib. It provides an easy way to create attractive and useful statistical graphs. This library makes us simple to create complex plots like heatmaps, violin plots, and pair plots. They are very helpful for exploring data. Seaborn also works really well with pandas letting us easily visualize data from DataFrames (Michael Waskom, 2012-2024).



Figure 9: Seaborn Logo

3.4.4. Evaluation

1. Scikit-learn

Scikit-learn (Sklearn) is a popular and useful library for machine learning in Python. It provides tools for tasks like classification, regression, clustering, and dimensionality reduction with a simple interface. The library is mostly written in Python and is built on top of NumPy, SciPy, and Matplotlib. It is reliable and easy to use for machine learning tasks (TutorialsPoint, 2024).



Figure 10: Scikit-learn Logo

3.4.5. Development Environment

1. Jupyter Notebook

The Jupyter Notebook is a web application for creating and sharing computational documents. We can write code in more than 40 programming languages, including Python. We can use big data tools like Apache Spark with Python, R, and Scala and also analyze data with tools like pandas, sci-kit learn and so on (Jupyter, 2024).



Figure 11: Jupyter Notebook Logo

2. VS Code

Visual Studio Code is a simple and lightweight code editor that supports tasks like debugging, running tasks, and version control. It provide the basic tools needed for quickly writing, building, and debugging code. The more advanced feature is available in full IDEs like Visual Studio (Microsoft, 2024).



Figure 12: VS Code Logo

4. Results

4.1. Results achieved (Evaluation)

The evaluation of both models was conducted using performance metrics such as R^2 score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Higher R^2 score and lower MAE, MSE and RMSE indicate better performance of the model.

4.1.1. Linear Regression Results

```
Linear Regression Evaluation Metrics:
=====================================
R² Score: 0.64
Mean Absolute Error (MAE): 0.45
Mean Squared Error (MSE): 0.38
Root Mean Squared Error (RMSE): 0.61
```

Figure 13: Regression model evaluation results

The results achieved by the Linear Regression model is shown in the figure above. With R^2 score as 0.64, MAE as 0.45, MSE as 0.38 and RMSE and 0.61, it indicates that the model reasonably well but it struggled to capture the complexity of the data.

4.1.2. ANN Results

```
ANN Evaluation Metrics:
=======================
R² Score: 0.72
Mean Absolute Error (MAE): 0.39
Mean Squared Error (MSE): 0.29
Root Mean Squared Error (RMSE): 0.54
```

Figure 14: ANN model evaluation results

The results achieved by the ANN model are shown in the figure above. With an R^2 score of 0.72, MAE of 0.39, MSE of 0.29, and RMSE of 0.54, it indicates that the model able to

capture the complexity of the data more effectively and was better performing than the Linear Regression model.

4.2. Model in Action

Both the Linear Regression and Artificial Neural Network (ANN) models were developed using PyTorch which is a popular deep learning framework. Both of them are explained below:

4.2.1. Linear Regression Model

1. Model Architecture

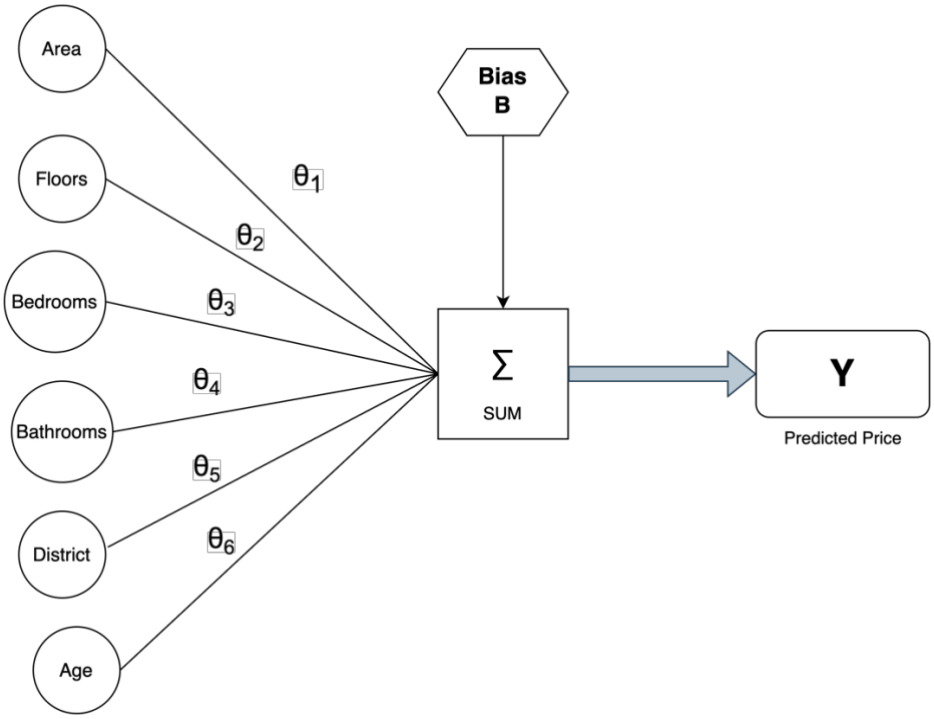


Figure 15: Regression model architecture

The Linear Regression model uses 6 features to predict house prices. Each feature has a weight/slope (θ1, θ2 ,…,θ6) that shows by what scale it affects the prediction. There is also a bias term (b) that adjusts the prediction up or down. These weights and the bias are combined to calculate the final predicted price.

2. Training

```
# Linear regression model
model = LinearRegression()

# Loss function for regression: Mean Squared Error (MSE)
criterion = nn.MSELoss()

# Optimizer Adam
optimizer = torch.optim.Adam(model.parameters(), lr=0.01)
```

Figure 16: Regression model criterion and optimizer code

The model is trained using the Mean Squared Error (MSE) as the loss function. It helps to measure how far the predictions are from the actual prices. The Adam optimizer is used for updating the model weights during training, with a learning rate of 0.01.

3. Loss

```
Epoch [10/150], Loss: 1.8748
Epoch [20/150], Loss: 0.9003
Epoch [30/150], Loss: 0.5229
Epoch [40/150], Loss: 0.4472
Epoch [50/150], Loss: 0.4286
Epoch [60/150], Loss: 0.4036
Epoch [70/150], Loss: 0.3854
Epoch [80/150], Loss: 0.3762
Epoch [90/150], Loss: 0.3699
Epoch [100/150], Loss: 0.3648
Epoch [110/150], Loss: 0.3613
Epoch [120/150], Loss: 0.3587
Epoch [130/150], Loss: 0.3568
Epoch [140/150], Loss: 0.3554
Epoch [150/150], Loss: 0.3542
```

Figure 17: Regression model training loss

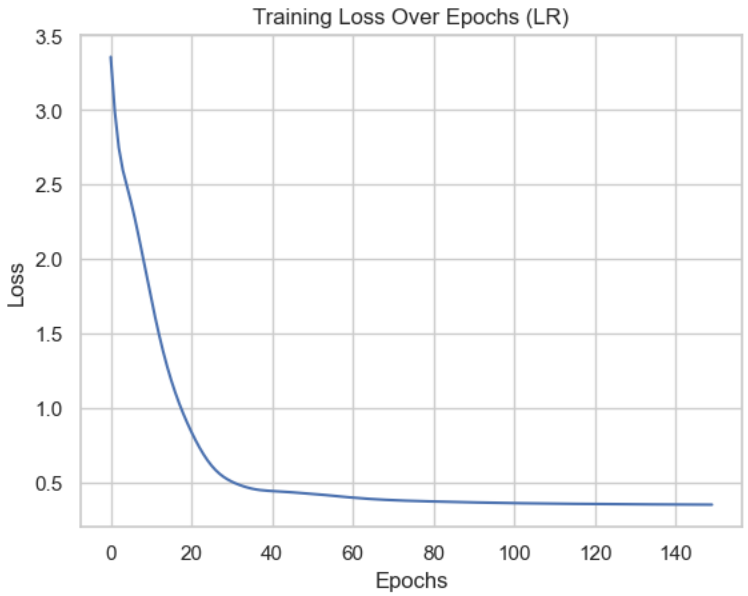


Figure 18: Regression model loss curve

The training progress is shown in the figure above with the loss curve after training. At the 10th epoch, the loss was 1.8748 and by the 150th epoch it was reduced to 0.3542. This consistent decrease in loss shows that the model improved steadily throughout the training process. The loss curve supports this visually. This indicates effective learning of the model.

4.2.2. ANN Model

1. Model Architecture

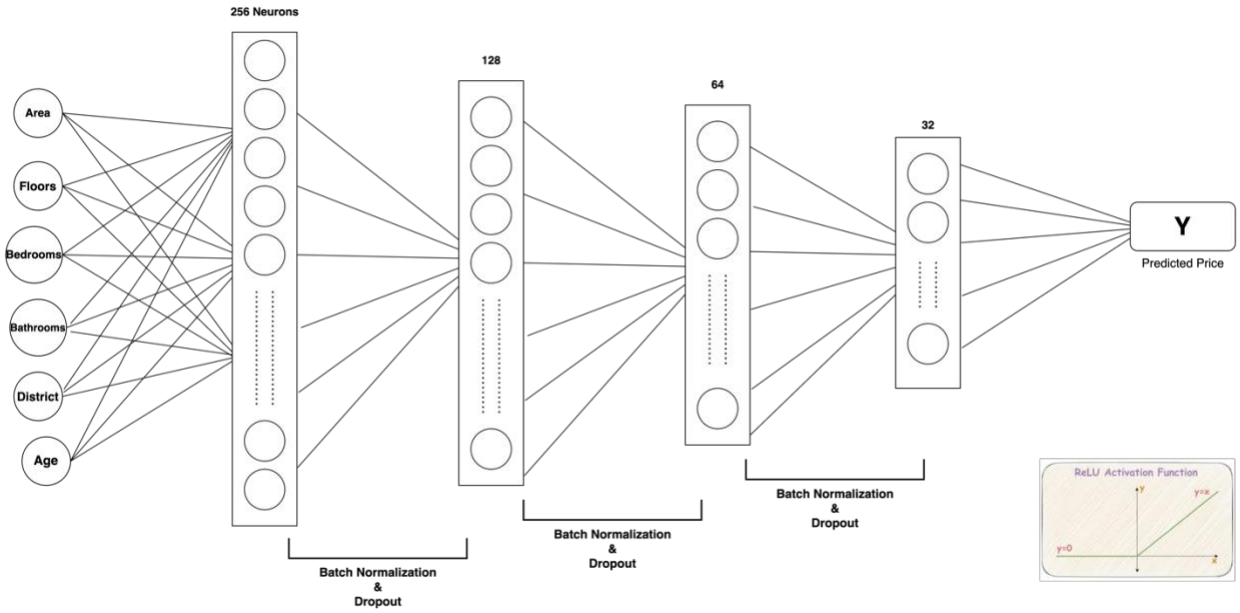


Figure 19: ANN model architecture

The Artificial Neural Network (ANN) model predicts house prices based on six features. It consists of four hidden layers with 256, 128, 64, and 32 neurons. After each layer, the data is batch normalized for stability and dropout is applied to prevent overfitting. The ReLU activation function is used throughout the network to capture non-linear relationships effectively. This architecture helps the model to learn complex patterns and make accurate predictions.

2. Training

```
# ANN model
model = ANN()

# Loss function for regression: Mean Squared Error (MSE)
criterion = nn.MSELoss()

# Optimizer Adam
optimizer = torch.optim.Adam(model.parameters(), lr=0.01)
```

Figure 20: ANN model criterion and optimizer code

The ANN model is also trained using the Mean Squared Error (MSE) loss function. It also uses the adam optimizer with learning rate of 0.01.

3. Loss

```
Epoch [5/50], Loss: 0.6331
Epoch [10/50], Loss: 0.5468
Epoch [15/50], Loss: 0.4782
Epoch [20/50], Loss: 0.4575
Epoch [25/50], Loss: 0.4255
Epoch [30/50], Loss: 0.4199
Epoch [35/50], Loss: 0.4165
Epoch [40/50], Loss: 0.4084
Epoch [45/50], Loss: 0.4024
Epoch [50/50], Loss: 0.4022
```

Figure 21: ANN model training loss

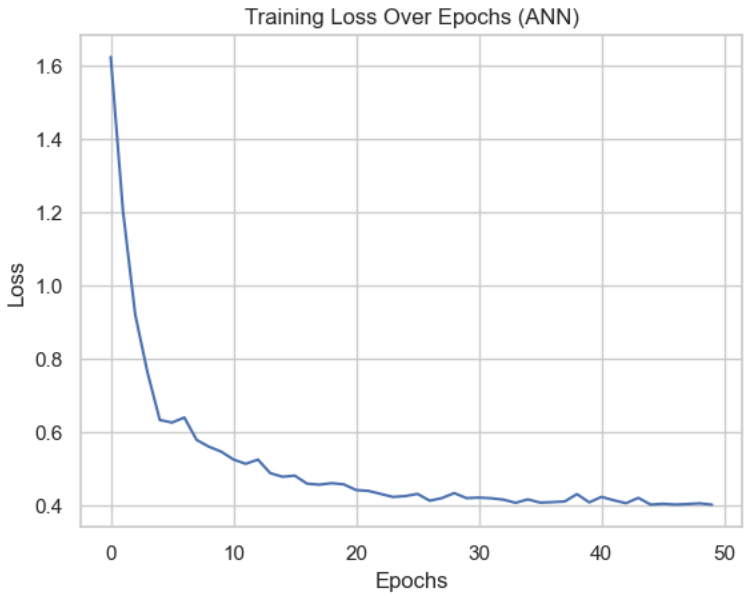


Figure 22: ANN model loss curve

The training progress for the ANN model is shown in the figure above with the loss curve after training. At the 5th epoch, the loss was 0.6331 and by the 50th epoch, it was reduced to 0.4022. This steady decrease in loss demonstrates that the model improved consistently during the training process. The loss curve visually confirms this progress, indicating effective learning by the ANN model.

4.3. Predictions

4.3.1. User Interface

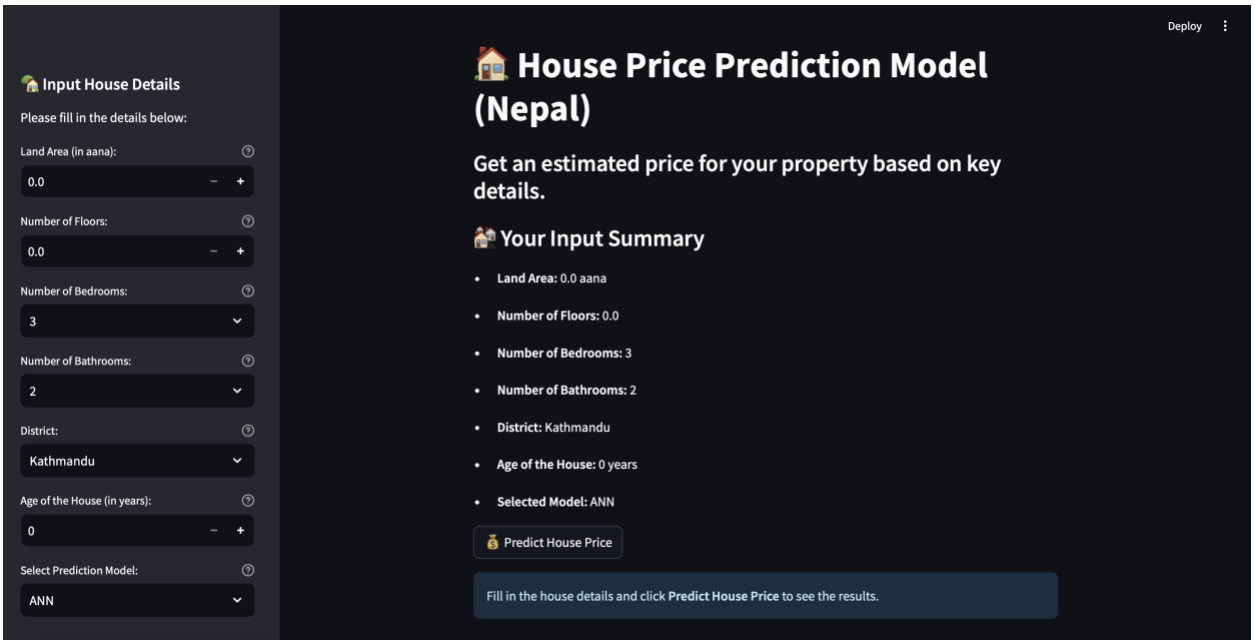


Figure 23: User Interface for prediction (Streamlit)

The user interface was developed in using streamlit. It is a powerful open source python framework. Streamlit is mostly used my ML and Data scientists for quick frontend and deployment of machine learning models.

4.3.2. Prediction Using Linear Regression Model



Figure 24: Prediction on linear model

Using the input values of land area as 5 aana, number of floors as 2, number of bedrooms as 2, number of bathrooms as 3, district as Lalitpur, and house age as 6 years, the Linear Regression model predicted a house price of 3.71 crores. This prediction is not that bad in the context of Nepal's real estate market.

4.3.3. Prediction Using ANN Model

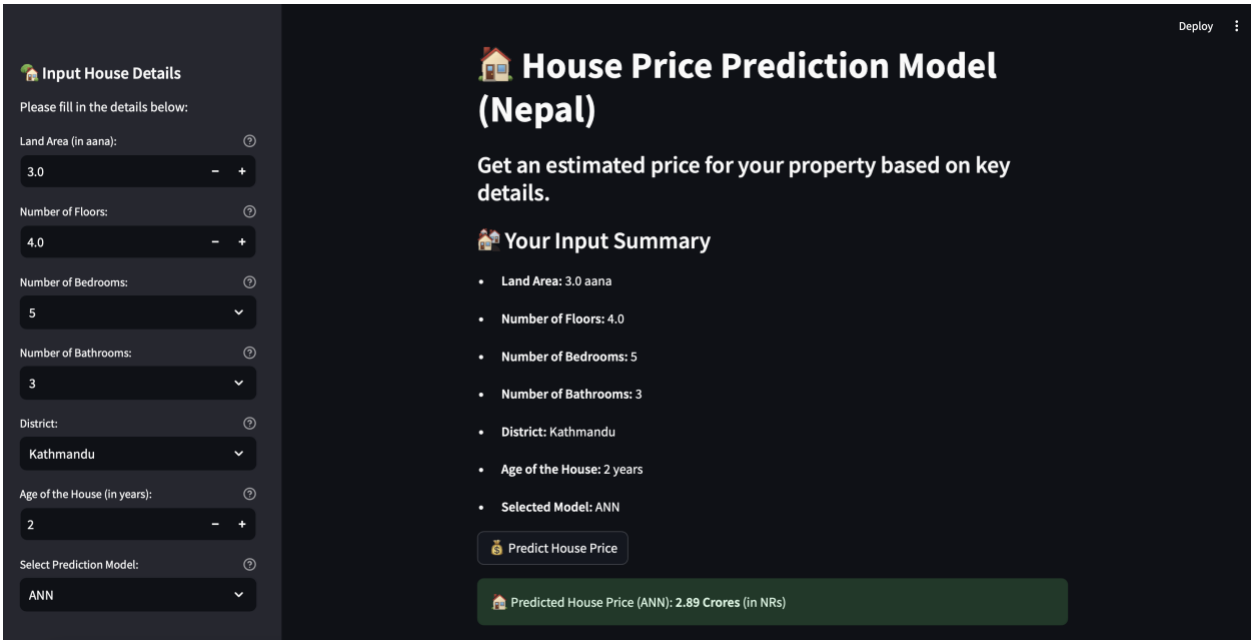


Figure 25: Prediction on ANN model

Similarly, using the input values of land area as 3 aana, number of floors as 4, number of bedrooms as 5, number of bathrooms as 3, district as Kathmandu, and house age as 2 years, the ANN model predicted a house price of 3.71 crores. This prediction is also quite accurate in the context of Nepal's real estate market.

5. Conclusion

The models used for predicting house prices, Linear Regression and ANN, have shown to be effective in providing predictions based on key property details. While the Linear Regression model performed good, the ANN model delivered better results, with improved accuracy in predicting prices. Both models performed well in showing the trends seen in the real estate market in Nepal.

This application really useful for simplifying the process of estimating house prices. It can be helpful for buyers, sellers, and real estate professionals by offering an easy way to get the property value and helping people make more informed decisions in a timely manner. This can also be a great help for better investments for future.

In terms of future improvements, adding additional features such as specific neighborhood data, distance from city center, extra facilities could improve the accuracy. We could also explore more advanced algorithms or advanced market data to keep predictions more aligned with evolving market conditions. These additions would make the tool more robust and valuable.